Gravity from Quantum Measurement

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Abstract

We propose a framework in which gravity can emerge from the physical backreaction of quantum measurement and the microstate dynamics of information acquisition. Standard results, including horizon scales and entropy bounds, follow from Landauer's principle at the Unruh temperature. We then present a concrete bilocal microstate construction as a case-study. Entangled pairs emitted from a localized source propagate into left and right Rindler wedges, where polarization measurement events produce discrete energy-momentum impulses and correlated classical bits. These microstates can be described geometrically using Aichelburg–Sexl pp-wave solutions, whose longitudinal shifts and transverse focusing imprint energy and information directly onto horizon geometry. This setup illustrates how individual measurement events contribute to horizon entropy, and how entanglement can be converted into spatial information, providing an operational realization of ER=EPR at the level of single microstates. The framework complements entropic and thermodynamic approaches by supplying a concrete, microphysical model linking quantum measurement, horizon dynamics, and emergent spacetime.

1 Introduction

Two landmark results establish that gravity and thermodynamics are deeply intertwined. Jacobson showed that the Einstein equations can be derived from thermodynamic identities applied to local Rindler horizons, with horizon entropy playing the role of the fundamental quantity from which spacetime dynamics emerge [1]. Verlinde extended this perspective, arguing that gravitational forces arise from entropy gradients on holographic screens, recasting gravity as an entropic rather than a fundamental interaction [2]. Both frameworks, however, operate at the level of bulk thermodynamics and leave open a prior question: what physical process underlies each individual entropy increment?

We argue that the answer is quantum measurement. Each act of information acquisition by a physical observer constitutes a discrete microstate event, and the collective backreaction of such events—together with the causal correlations they induce—gives rise to gravitational effects. The argument proceeds in three steps. First, any observation is mediated by a physical interaction: information transfer necessarily involves an exchange of energy and momentum between the observed system and the detector, a discrete impulse. Second, by the equivalence principle, this impulse is locally indistinguishable from the effect of a gravitational field: no local measurement can distinguish the recoil associated with a quantum measurement event from free fall in a gravitational potential. Third, when such impulses are modelled geometrically—via Aichelburg–Sexl (AS) pp-wave solutions, as we show—they imprint energy and information directly onto horizon geometry,

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and coarse-graining over the resulting ensemble recovers standard horizon thermodynamics.

A unifying structure runs through all three of these relations, and it is worth making explicit before proceeding. In each case, the relation between two adjacent concepts has the same logical character: every instance of the more abstract concept is grounded in an instance of the more physical one, but not every instance of the more physical concept gives rise to the more abstract one *for a given observer*. The gap in each case is filled by subjectivity. Quantum state reduction is always backed by an observation—it just is a Bayesian update on the observer's part—but an observation that carries no new information for that observer produces no state reduction. Equally, every observation is backed by an energy-momentum transfer, but not every such transfer constitutes an observation for you: an interaction involving a different observer leaves your state unchanged. The Unruh effect suggests the same structure extends to the equivalence principle: an accelerating observer perceives a thermal bath and attributes it to a gravitational field, while an inertial observer sees only vacuum. Whether an acceleration counts as gravity is perspectival in precisely the same sense. In each case the relation is an identification, not a causal link, and the identification is observer-relative.

This stands in sharp contrast to the Penrose–Diósi objective collapse programme [3, 4], which postulates that gravity *causes* quantum state reduction as an objective, observer-independent physical process. Figure 1 summarises this contrast. The three solid arrows represent the perspectival grounding relations described above; the single dashed arrow represents the Penrose–Diósi causal postulate. It is the only relation in the diagram claimed to be objective and observer-independent, and it is the only one we do not adopt.

$$\begin{tikzcd}
\text{Gravitational field} \arrow[r, dashed, "\text{Penrose–Diósi}"] & \text{Quantum state reduction} \\
\text{Acceleration} \arrow[u, "\text{equivalence principle}"] \arrow[r, "\text{energy–momentum transfer}"'] & \text{Observation} \arrow[u, "\text{Bayesian update}"']
\end{tikzcd}$$

Figure 1: Solid arrows denote substrate relations: $A \to B$ means every instance of B is physically realised by an instance of A, for a given observer. The dashed arrow denotes the Penrose–Diósi postulate, where the relation is claimed to hold objectively rather than observer-relatively.

To make this concrete, we build on the microstate construction of [5], in which bilocal driving sources generate entangled Rindler excitations that realise individual terms of the Unruh thermal ensemble. The present paper applies and extends that construction. Entangled pairs emitted from a localised source propagate into left and right Rindler wedges; polarisation measurement events produce discrete energy-momentum impulses and correlated classical bits. We show that each such microstate carries a calculable gravitational imprint, modelled by an AS pp-wave solution, and that coarse-graining over the ensemble recovers standard horizon thermodynamics. This construction also provides an operational realisation of ER=EPR at the level of individual microstates: entanglement is converted into geometric information, and each measurement event contributes a discrete, calculable increment to horizon entropy.

This approach both complements and sharpens the thermodynamic gravity programme of Jacobson and Verlinde. Jacobson's derivation of the Einstein equations from horizon thermodynamics and Verlinde's entropic force framework both operate at the level of

bulk averages, leaving open the question of what physical process underlies each entropy increment. The present construction supplies that microstate realisation: each entropy increment corresponds to a physically mediated measurement event with a calculable energy-momentum transfer. Thermality is emergent rather than intrinsic, arising from coarse-graining over an ensemble of source-driven microstates, each of which admits a definite, non-thermal description in the global Minkowski frame. The thermodynamic results of both programmes are recovered as a consequence of this coarse-graining, rather than assumed as a starting point.

[Section map paragraph todo]

2 Motivating Heuristics

The central observation is simple. Landauer's principle associates a minimal energy cost with the acquisition of information [6]. For an accelerated observer, the relevant temperature is the Unruh temperature [7]. Substituting one into the other eliminates temperature entirely, yielding a frame-independent relation between information acquisition, energy transfer, and acceleration (a version of the BHT first law). This is not merely an algebraic convenience: it is a concrete dictionary between the thermal description natural to a Rindler observer and the non-thermal global Minkowski description, with the apparent thermality of the vacuum emerging as a consequence of the observer's acceleration rather than as a fundamental property of the state. In this way we derive a sequence of standard results in black hole thermodynamics.

2.1 The Unruh-Landauer Relation

Landauer's principle states that the acquisition of ΔS bits of information by a physical observer requires a minimum energy transfer

$$\Delta E \geq k_B T \, \Delta S. \tag{1}$$

For an observer undergoing proper acceleration a, the Unruh effect associates a temperature

$$T = \frac{\hbar a}{2\pi k_B c} \tag{2}$$

with the local vacuum. Substituting into Landauer's bound eliminates $k_B T$, yielding

$$\boxed{\Delta E \geq \frac{\hbar a}{2\pi c} \, \Delta S.} \tag{3}$$

Temperature has dropped out entirely: equation (3) is a frame-independent relation between energy, acceleration, and information, with no reference to any thermal description. It can be read as the first law of black hole thermodynamics derived from below — from the physical cost of information acquisition — rather than postulated as a thermodynamic identity. In what follows, all standard results emerge as consequences of this single relation. We note that Landauer's principle is conventionally stated for information erasure rather than acquisition. Here we apply it to acquisition, which is the physically relevant process for an observer registering a measurement outcome. The bound has the same form in both cases, and we proceed with the understanding that this is a heuristic application whose precise justification is deferred to the microstate construction of Section **??**.

2.2 Momentum Transfer and Acceleration

The energy transfer ΔE implied by equation (3) is carried by a physical quantum — matter or radiation — mediating the measurement interaction. We refer to this as the

thrust quantum. For a relativistic carrier, the associated momentum transfer is

$$\Delta p = \frac{\Delta E}{c}. \tag{4}$$

Let the interaction occur over a spatial scale r, so that the interaction time is $\Delta t \sim 2r/c$. For a detector of mass m_d, Newton's second law in impulse form gives

$$a = \frac{\Delta p}{m_d \Delta t} = \frac{\Delta E}{2 m_d r}. \tag{5}$$

The factor of 2 in the interaction time is not an independent assumption: it will be fixed self-consistently by requiring that the characteristic interaction scale reproduces the Schwarzschild radius, as shown in the next subsection.

2.3 The Schwarzschild Radius

The thrust quantum carries energy ΔE and produces an acceleration a on the detector. Identifying this with the Newtonian gravitational acceleration due to a mass m_t at distance r.

$$a = \frac{G m_t}{r^2}, \tag{6}$$

and substituting equation (5) with $\Delta E = m_t c^2$, we obtain

$$\frac{m_t c^2}{2 m_d r} = \frac{G m_t}{r^2}. \tag{7}$$

Solving for r:

$$r = \frac{2 G m_d}{c^2} \equiv r_s, \tag{8}$$

which is precisely the Schwarzschild radius of the detector mass m_d. This fixes the factor of 2 in the interaction time retroactively: the choice $\Delta r = 2r/c$ is the unique one consistent with Newtonian gravity and the correct Schwarzschild scale.

2.4 Surface Gravity

Evaluating the acceleration at $r = r_s$ gives the surface gravity of the detector:

$$\kappa = \frac{G m_d}{r_s^2} = \frac{c^4}{4 G m_d}, \tag{9}$$

in agreement with the standard Schwarzschild result. This is the acceleration that appears in the Unruh temperature associated with the detector's horizon, $T = \hbar\kappa / 2\pi k_B c$, providing an internal consistency check on the framework.

2.5 Bekenstein-Hawking Entropy

Returning to the Unruh-Landauer relation (3) and substituting the surface gravity κ for a, the entropy increment associated with an energy increment $\Delta E = c^2 \Delta m_d$ is

$$\Delta S \leq \frac{2\pi c}{\hbar \kappa} \Delta E = \frac{8\pi G m_d}{\hbar c^3} \cdot c^2 \Delta m_d. \tag{10}$$

Expressing this in terms of the change in Schwarzschild area $\Delta A = 32\pi G^2 m_d \Delta m_d / c^4$:

$$\Delta S = \frac{\Delta A}{4 \ell_P^2}, \qquad \ell_P^2 = \frac{\hbar G}{c^3}, \tag{11}$$

which is the Bekenstein-Hawking area law. Each measurement event contributes a discrete entropy increment proportional to the area imprinted on the horizon by the associated energy-momentum transfer.

2.6 Summary

Beginning from the single relation (3), and using only Newton's law and the equivalence principle, we have recovered the Schwarzschild radius, the surface gravity, and the Bekenstein-Hawking area law as consequences of the observation=acceleration identification. These are standard results, but their derivation here from the physical cost of information acquisition rather than from thermodynamic postulates is the point. The thrust quantum introduced in this section — the physical carrier of the measurement-induced impulse — will be given a precise microstate realization in the sections that follow, where it is identified with an individual photon modeled as an Aichelburg-Sexl pp-wave.

3 Bilocal Microstate Construction

The heuristic arguments of Section 2 establish a relationship between information acquisition, acceleration, and gravitational effects, but leave open the question of what a single microstate looks like concretely. In standard treatments, the Unruh effect is derived by coupling a detector to a pre-existing acceleration and reading off a thermal response from the vacuum. Following [5], we invert this logic: we place an interaction in the past, where a source injects non-thermal entangled excitations into the field, and show that the observer's accelerated response is driven by these source-induced excitations. The thermal ensemble then emerges as a coarse-grained aggregate of such microstates, rather than being a primitive property of the vacuum.

3.1 The Paz–Lester–Wright Setup

We adopt an Alice-and-Bob style naming convention. Three observers are placed at spacetime positions $(t, z) = (0, 1)$, $(0, -1)$, and $(-1, 0)$, named *Wright* (R), *Lester* (L), and *Paz* (P) respectively — a mnemonic for Right, Left, and Past. Paz serves as the common causal ancestor of Wright and Lester, see Figure 2.

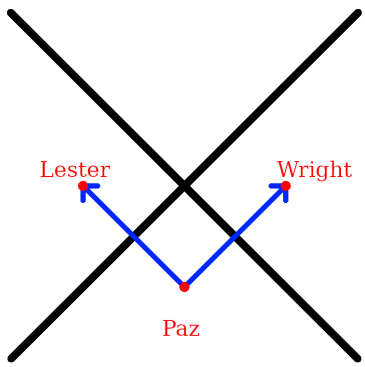

Figure 2: A source in the past, Paz(P), emits an entangled particle pair that accelerates Wright(R) (to the right) and Lester(L) (to the left).

Paz emits an entangled pair of spin-0 massless scalar quanta[1] with momenta $\pm k$ and Rindler frequency $\omega = \omega_k$. The momenta are perfectly balanced, so the source exhibits no back reaction in this minimal model — a simplification discussed further in Section **??**. The two quanta propagate into the left and right Rindler wedges, where Lester and Wright detect them respectively.

[1]Strictly speaking, these are spin-0 massless scalar quanta; the polarization language used elsewhere is heuristic.

The state prepared by this emission is a controlled realization of a single term in the Unruh thermal ensemble:

$$|\Psi_\omega\rangle_M = |1_\omega\rangle_L \, |1_\omega\rangle_R + \mathcal{O}(\text{higher order}), \tag{12}$$

rather than a superposition sampled from a thermal distribution. Wright's detection is therefore guaranteed, and is modeled as a PVM projection selecting a specific component of the thermal ensemble. The source does not contradict the thermal interpretation; it describes a specific microstate consistent with the broader ensemble.

3.2 The Bilocal Source

Paz's emission is modeled as a bilocal driving source coupling to the quantum field ϕ:

$$J(x, y) = \lambda \, f_L(x) \, f_R(y), \tag{13}$$

with the sourced Lagrangian

$$\begin{aligned} \mathcal{L}_{\text{sourced}} &= \mathcal{L}_{\text{free}} + \frac{1}{2}\lambda \int f_L(x) \, f_R(y) \, \phi_+(x) \, \phi_+(y) \; + \text{h.c.} \\ &= \mathcal{L}_{\text{free}} + \frac{1}{2}\lambda \left(q^{L\dagger}_{-k} q^{R\dagger}_{k} + q^{L}_{-k} q^{R}_{k} \right), \end{aligned} \tag{14}$$

where λ is a coupling constant, f_L and f_R are normalized mode profiles supported on the left and right wedges respectively, ϕ_+ is the positive-frequency part of ϕ, and $q^{L\dagger}_{-k}$, $q^{R\dagger}_{k}$ are creation operators for the corresponding modes. The last expression in equation (14) is a squeezing interaction at frequency ω: the bilocal source creates entangled Rindler excitations in both wedges simultaneously, in direct analogy with quadrature squeezed light generation in quantum optics [**?**]. This is a quadratic generalization of the standard linear source in QFT [**?**, **?**], with the product structure reflecting the paired nature of the excitations.

When f_L and f_R are chosen to be the Rindler modes l_{-k} and r_k, smearing with ϕ_+ projects them onto the corresponding Unruh modes μ^L_{-k} and μ^R_k, so that $q^{L\dagger}_{-k}$ and $q^{R\dagger}_{k}$ become the Unruh operators $c^{L\dagger}_{-k}$ and $c^{R\dagger}_{k}$. The resulting state is precisely equation (12). Table **??** summarises the correspondence between mode profiles, operators, and states.

The connection to the standard Unruh vacuum is made explicit by writing the Minkowski vacuum in the Rindler basis:

$$|0\rangle_M = \prod_\omega \frac{1}{\cosh \theta_k} \sum_n \tanh^n \theta_k \, |n_\omega\rangle_L \, |n_\omega\rangle_R. \tag{15}$$

The bilocal source selects a single term from this sum at fixed ω, replacing the full thermal superposition with a definite, non-thermal microstate. Apparent thermality arises from coarse-graining over many such microstates — from effective ignorance of the source's detailed structure — not from any intrinsic property of the vacuum.

3.3 Entanglement as Spatial Information: ER=EPR at the Microstate Level

Each measurement produces a polarization outcome — H or V — at each wedge, perfectly correlated between Lester and Wright. But consider a different question: not *what* was observed, but *where*. The question "who observed H?" has a binary answer: Lester (left wedge) or Wright (right wedge). This is a spatial bit — a bit that encodes not a polarization state but a location in space.

The entangled pair has therefore converted quantum information — polarization entanglement — into classical spatial information — the left/right correlation. The Rindler

horizon is the substrate on which this information is physically encoded: the two wedges are the two sides of the horizon, and the correlated measurement outcomes are the horizon's microstates. This is an operational realisation of ER=EPR at the level of individual microstates, with the entanglement between Lester and Wright's outcomes playing the role of the Einstein–Rosen bridge connecting their respective wedges.

4 Gravitational Imprint: the pp-Wave Realisation

Each measurement event in the bilocal construction involves a photon transferring energy and momentum to an accelerated detector. We now model the gravitational imprint of this transfer by identifying each photon with an Aichelburg–Sexl (AS) pp-wave solution in general relativity [**?**]. The AS geometry is an exact solution describing the gravitational field of an ultrarelativistic point mass, and provides the natural geometric language for a single-photon microstate.

4.1 The AS pp-Wave and Its Effects

Let $u = t - z$ and $v = t + z$, where the photon propagates along the null plane $u = 0$, z is the longitudinal direction, and (x, y) are transverse coordinates. Test particles at fixed transverse positions (x, y) experience two effects as the shock front crosses $u = 0$:

1. **Longitudinal shift.** The v coordinate is displaced by

$$\Delta v \propto -\ln r^2, \qquad r^2 = x^2 + y^2. \tag{16}$$

   This shift is lightlike: no proper time elapses for a particle crossing the shock, but it is displaced along the null direction. Near the axis ($r \to 0$) the shift diverges, encoding the concentration of energy along the photon's trajectory.

2. **Transverse focusing.** The gradient of the longitudinal shift produces a transverse velocity

$$\Delta \mathbf{v}_\perp \propto \nabla \ln r \propto \frac{1}{r}, \tag{17}$$

   directed toward the axis. Test particles are pulled toward the centre of the wave, regardless of their initial transverse separation.

4.2 The Longitudinal Shift Makes Space

The longitudinal shift deserves closer attention, because it carries a meaning beyond a mere coordinate displacement. The photon emitted by Paz propagates along the null direction; as it crosses the Rindler horizon, the AS shift displaces the future light cone of every test particle it passes. For the photon traveling into the right wedge, the shift pushes the right wedge's geometry in the $+v$ direction; for the entangled partner traveling into the left wedge, the shift pushes the left wedge in the $-v$ direction. The two wedges are displaced *away from each other* along the longitudinal null direction.

This is not merely a kinematic effect. The transverse plane of the pp-wave is identified with horizon area elements $dA = dx\,dy$, and the longitudinal shift creates a new null interval between the two wedges — a region of spacetime that did not exist, in the sense of being causally accessible, before the photon passed. The bit of spatial information encoded in the left/right measurement outcome — "who observed H?" — refers to precisely this separation. The measurement event does not merely *record* a spatial distinction; through the gravitational backreaction of the photon, it *creates* the spatial separation that the bit describes.

Put differently: the same physical event — a photon crossing a horizon — simultaneously encodes a classical bit and, via the AS longitudinal shift, generates the spatial structure that bit refers to. Space is not a pre-existing backdrop onto which information is written; it is produced, increment by increment, by the gravitational imprint of each measurement event. This is the microstate-level content of emergent space: each detection event contributes both an entropy increment and a spatial increment, and the two are the same thing described in different languages.

4.3 Horizon Entropy from Individual Microstates

Each detection event contributes a discrete entropy increment via the Bekenstein–Hawking relation derived in Section 2:

$$\Delta S = \frac{\Delta A}{4\ell_P^2}, \qquad \ell_P^2 = \frac{\hbar G}{c^3}, \tag{18}$$

where ΔA is the area element imprinted on the horizon by the transverse focusing of the pp-wave. Horizon entropy is therefore not a bulk thermodynamic quantity assigned to the horizon as a whole, but a sum of discrete, calculable contributions from individual measurement-induced interactions. The transverse focusing and the longitudinal shift are two aspects of the same microstate: focusing encodes the entropy increment in the area; the longitudinal shift encodes the spatial increment in the geometry.

4.4 Coarse-Graining and the Emergence of Thermality

In a steady-state scenario, Paz continuously emits entangled pairs, and Lester and Wright continuously detect them. Each detection is an AS pp-wave microstate with a definite, non-thermal description in the global Minkowski frame: a specific photon energy, a specific transverse position, a specific longitudinal shift. Coarse-graining over this ensemble recovers the Unruh thermal spectrum at temperature $T = a/2\pi$, consistent with the vacuum decomposition of equation (15). The thermodynamic description of the horizon emerges from this averaging; it is not a feature of the individual microstates.

This completes the microstate programme. Each entropy increment in the Bekenstein–Hawking area law corresponds to a single AS pp-wave interaction. Each pp-wave interaction corresponds to a single measurement event in the bilocal construction. Each measurement event both encodes a spatial bit and generates the spatial separation that bit refers to. The gravitational, thermodynamic, information-theoretic, and geometric descriptions of the horizon are unified at the level of individual microstates, and thermality, entropy, and space itself emerge together from the coarse-grained aggregate of these events.

5 Conclusion

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